Machine Learning Algorithms Cheat Sheet

Supervised Learning Algorithms

Linear Regression

How it works: Predicts a continuous target using a linear relationship between input and output.
Formula: $y = \beta 0 + \beta 1x1 + ... + \beta nxn + \varepsilon$
Advantages:
- Easy to implement and interpret.
- Computationally efficient.
Disadvantages:
- Assumes linearity.
- Sensitive to outliers.
Comparison:
- Simpler than Ridge/Lasso.
- Less robust than tree-based models for nonlinear data.

Ridge Regression

How it works: Adds L2 regularization to penalize large coefficients.
Formula: $Cost = \Sigma(yi - \hat{y}i)^2 + \lambda \Sigma\beta j^2$
Advantages:
- Prevents overfitting.
- Works well with multicollinearity.
Disadvantages:
- Doesn’t perform feature selection.
Comparison:
- Better generalization than Linear Regression.
- Unlike Lasso, doesn't reduce coefficients to zero.

Lasso Regression

How it works: Uses L1 regularization to reduce less important feature weights to zero.
Formula: $Cost = \Sigma(yi - \hat{y}i)^2 + \lambda \Sigma|\beta j|$
Advantages:
- Performs feature selection.
Disadvantages:
- Can arbitrarily select one of several correlated features.
Comparison:

- More interpretable than Ridge.
- Less stable with highly correlated predictors.

K-Nearest Neighbors (K-NN)

How it works: Predicts label based on the majority (or mean) of K closest data points.
Advantages:
- Simple and non-parametric.
Disadvantages:
- Slow for large datasets.
- Sensitive to irrelevant features and feature scaling.
Comparison:
- More intuitive than probabilistic models.
- Worse scalability than SVM or Logistic Regression.

Naive Bayes

How it works: Based on Bayes Theorem with an assumption of feature independence.
Formula: $P(y|X) = (P(X|y)P(y))/P(X)$
Advantages:
- Fast and works well with high-dimensional data.
Disadvantages:
- Strong independence assumption rarely holds.
Comparison:
- Fast like KNN but more scalable.
- Less flexible than tree models.

Support Vector Machine (SVM)

How it works: Finds the optimal hyperplane that maximizes the margin between classes.
Advantages:
- Effective in high-dimensional spaces.
- Can use kernel trick for non-linear separation.
Disadvantages:
- Computationally intensive for large datasets.
Comparison:
- More accurate than logistic regression in complex spaces.
- Slower than Naive Bayes or Logistic Regression.

Decision Tree

How it works: Splits data recursively based on feature thresholds to minimize impurity.
Impurity Metrics:
- Gini Index
- Entropy
Advantages:
- Easy to visualize and interpret.
- Handles both numerical and categorical data.

Disadvantages:
- Prone to overfitting.
Comparison:
- Less stable than ensemble methods (Random Forest).
- More interpretable than SVM.

Random Forest

How it works: Ensemble of decision trees using bagging and feature randomness.
Advantages:
- Reduces overfitting.
- Works well for both classification and regression.
Disadvantages:
- Less interpretable.
Comparison:
- Better generalization than single trees.
- Slower than simpler models like Logistic Regression.

Logistic Regression

How it works: Models the probability of a class using the logistic function.
Formula: $P(y=1|X) = 1 / (1 + e^{-(\beta 0 + \beta 1x1 + ... + \beta nxn)})$
Advantages:
- Probabilistic interpretation.
- Works well when linear boundary suffices.
Disadvantages:
- Can underperform on non-linear data.
Comparison:
- Better interpretability than SVM.
- Less powerful for complex boundaries.

Unsupervised Learning Algorithm

K-Means Clustering

How it works: Assigns data to k clusters by minimizing intra-cluster variance.
Objective Function: $\Sigma(i=1 \text{ to } k) \Sigma(x \in Ci) ||x - \mu i||^2$
Advantages:
- Simple and scalable.
Disadvantages:
- Requires prior knowledge of K.
- Sensitive to initialization and outliers.
Comparison:
- Faster than any other clustering techniques.

Underfitting vs Overfitting

Underfitting:
- High bias, low variance.
- Poor performance on both training and test sets.
- Model too simple.

Overfitting:
- Low bias, high variance.
- Excellent performance on training, poor on test set.
- Model too complex.

Good Fit:
- Balanced bias-variance tradeoff.
- Performs well on both training and unseen data.

Cross-Validation

What: Technique to validate model on different subsets of data.
Types:
- K-Fold Cross-Validation: Splits data into k subsets; each used once as test set.

Hyperparameter Tuning Techniques

Grid Search CV

Exhaustively searches through a specified parameter grid.
Pros: Thorough and easy to parallelize.
Cons: Computationally expensive.

Random Search CV

Samples random combinations of parameters.
Pros: Faster, good for large hyperparameter spaces.
Cons: May miss optimal combination.